

Fundamentals of Information Systems Security

Chapter 7 Quiz Answer Key

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Section A: Multiple Choice (40 points)

1. **A** — Software that performs actions contrary to the user's intentions
2. **B** — Virus
3. **B** — Malware that modifies or replaces programs to hide compromise
4. **C** — Ransomware
5. **A** — Viruses require user action; worms self-propagate
6. **C** — Phishing
7. **A** — A network of compromised computers controlled by an attacker
8. **B** — Denial-of-Service (DoS)
9. **C** — Reconnaissance
10. **A** — A decoy system designed to attract attackers

Section B: True/False (20 points)

1. **True** — Trojans do not self-replicate
2. **True** — Polymorphic viruses change appearance
3. **False** — Spam accounts for 70-90
4. **True** — Logic bombs execute when conditions are met
5. **False** — Keyloggers can also capture screenshots
6. **True** — SQL injection is an injection attack
7. **False** — Defense-in-depth uses multiple layers
8. **True** — Zero-day vulnerabilities have no patch
9. **False** — Spear phishing is highly targeted
10. **True** — War driving locates wireless networks

Section C: Short Answer (40 points)

1. Virus, Worm, and Trojan (10 points)

- **Virus:** Requires user action to execute and spread; attaches to legitimate programs; self-replicates
- **Worm:** Self-contained, self-replicating; spreads across networks without user interaction; exploits vulnerabilities
- **Trojan Horse:** Disguised as legitimate software; does NOT self-replicate; requires user to execute; spreads through social engineering

2. Five Phases of Cyber Attack (10 points)

- (a) **Reconnaissance:** Passive information gathering, OSINT
- (b) **Scanning:** Active network probing, port scanning, vulnerability mapping
- (c) **Gaining Access:** Vulnerability exploitation, privilege escalation
- (d) **Maintaining Access:** Backdoor installation, persistent malware
- (e) **Covering Tracks:** Log manipulation, evidence removal

3. Defense-in-Depth (10 points)

- **Definition:** Multiple layered defensive mechanisms; if one fails, another steps in
- **Layers:** Physical security, network security (firewalls, IDS/IPS), host security (antivirus, EDR), application security (secure coding, WAF), data security (encryption, DLP), user security (awareness training, MFA), incident response

4. Phishing, Spear Phishing, and Pharming (10 points)

- **Phishing:** Mass email attacks impersonating trusted entities to steal credentials
- **Spear Phishing:** Highly targeted phishing using victim-specific information
- **Pharming:** Redirects users from legitimate sites to fake ones by poisoning DNS; more dangerous because it doesn't require user to click a link